ORDERGUARD AI

Business Memo

A cost-aware approach to reducing Cash-on-Delivery Return to Origin (RTO)

TO	Founder	**DATE**	31 October 2026
FROM	OrderGuard AI Project Team	**SUBJECT**	COD RTO prevention strategy

Executive recommendation

Adopt a **graduated, review-first policy** rather than automatically cancelling every order with an elevated risk score. Use the model to estimate RTO likelihood, deterministic checks to flag incomplete or suspicious order details, and configurable thresholds to route orders to the least restrictive intervention justified. Reserve **HOLD_REVIEW** for the highest-risk cases or multiple serious validation failures. A hold is a manual-review state—not an automatic cancellation.

This balances Operations' need to reduce reverse-logistics costs with Marketing's concern that rejecting genuine first-time buyers wastes acquisition spend and damages trust. The current model is useful for prioritisation, but its precision is not strong enough to justify automatic cancellation.

What the data says

Finding	Observed result	Why it matters
Historical outcomes	2,018 records: 1,444 delivered, 573 RTO and 1 in transit. Among completed orders, RTO was about 28.4%.	The dataset reflects a material RTO problem, close to the brief's ~28% baseline.
Data quality	4 missing city values, 12 missing pincodes, 30 duplicate order IDs and 41 addresses recorded as ".".	Incomplete or malformed inputs can undermine model predictions; validate before routing.
Model comparison	Chronological holdout: Random Forest F1 55.0%, precision 48.5%, recall 63.5%, ROC-AUC 0.729; Logistic Regression F1 51.8%.	At this operating point, roughly half of positive RTO flags are false positives. Avoid score-only cancellation.

Surprise: the dataset contains obvious data-quality defects alongside outcome labels. A model cannot reliably compensate for missing or malformed order information. Input-quality checks therefore belong in the decision flow, not just in data preparation.

Economics and the Operations–Marketing trade-off

The stated direct cost of an RTO is approximately ■**169.40**: ■65 forward shipping + ■65 return shipping + ■25 packaging/handling + (8% × ■180) expected damage cost. This excludes additional support costs and the opportunity cost of delayed inventory.

The brief reports prepaid RTO of ~4% versus COD RTO of ~28%. The 24-percentage-point difference implies an indicative expected RTO-cost reduction of **■40.66 per order shifted from COD to prepaid** (0.24 × ■169.40), before discount and conversion effects. For a ■649 order, a 5% discount cap is ■32.45. The implemented calculator therefore caps the offer at the lower of ■40.66 and ■32.45: a maximum indicative offer of **■32.45**. This is a ceiling, not a reason to discount every order.

A genuine delivered ■649 order contributes about **■402.38** at the stated 62% gross margin, before acquisition and other costs. The brief separately notes ■300+ acquisition cost for new customers. Blanket cancellation can therefore destroy meaningful value, especially for first-time buyers. Prefer confirmation or a prepaid nudge where these can reduce RTO exposure without forfeiting the sale.

Recommended operating policy

Signal / risk band	Action	Rationale
Low predicted risk; no serious validation issue	**SHIP_AS_IS**	Avoid adding friction to orders that look reliable.
At/above confirmation threshold or moderate validation concern	**CONFIRM_ORDER**	Seek additional customer commitment; a score alone is not proof of fraud.
At/above prepaid threshold	**NUDGE_PREPAID**	Offer a capped discount only where expected avoided RTO cost supports it.
At/above hold threshold or multiple high-severity validation failures	**HOLD_REVIEW**	Pause fulfilment for human review. Do not automatically cancel solely on model score.

The prototype's initial configurable thresholds are 30% (confirmation), 50% (prepaid nudge) and 85% (hold for review). These are starting settings, not statistically optimised cut-offs. Tune them using a cost-sensitive validation exercise and review them with Operations and Marketing. The settings page allows non-technical operators to adjust thresholds.

What would this have done to last month's orders?

A defensible order-by-order counterfactual has **not yet been calculated**. The model metrics describe a chronological holdout, not how many historical orders would fall into each live decision bucket. We should not claim specific avoided RTOs, cancelled genuine orders or net rupee savings until every historical order has been rescored with the saved model, configured thresholds and validation logic. That replay should join predictions back to actual delivery labels and produce the four-bucket and cost summaries.

Risks, monitoring and next two weeks

Key risks: false positives may inconvenience genuine buyers; data drift may change patterns by city, courier or acquisition source; incomplete addresses may trigger unnecessary review; and discounts may reduce margin without enough prepaid conversions. Monitor action rates by first-time/repeat customer, city, courier and acquisition source. Track eventual delivery outcomes, false-positive rate, RTO rate, prepaid conversion, discount cost, review backlog and net contribution—not just model accuracy.

Next two weeks: (1) run the historical policy replay and publish bucket counts and rupee impact; (2) collect delivery outcomes for evaluated orders; (3) calibrate thresholds using false-positive and false-negative costs, with a protected first-order cohort; (4) replace prototype JSON storage with a transactional database and add audit logging; and (5) reconcile dashboard figures against the order ledger.

> **Decision requested:** Approve a controlled pilot of the graduated policy, with manual review for the highest-risk orders and no automatic cancellation based solely on the current model. Expand automation only after replay and live monitoring demonstrate positive net contribution without unacceptable loss of genuine orders.

Basis and limitations: Prepared from the take-home assignment brief and the project's current dataset/model analysis. Business assumptions (COD share, RTO rates, cost components, margin and acquisition-cost context) come from the brief. Dataset counts and model metrics are project analysis results. No historical policy replay or live conversion experiment is represented as completed. WhatsApp execution is not included in the current prototype; confirmation/prepaid actions are recommendations only.